

Railway Crew Management System (RCMS)

Product Requirements Document (PRD)

Version

1.0

Project Overview

The Railway Crew Management System (RCMS) digitizes railway crew operations by importing operational Excel exports from an existing CMS, storing them in PostgreSQL, and providing secure dashboards, reports, and historical tracking.

Objectives

Centralized crew database, Excel import, duty history, dashboards, reporting, role-based security, audit logs, and future CMS automation.

Completed Backend

FastAPI setup, PostgreSQL, SQLAlchemy, Alembic, JWT Authentication, Role-Based Access Control, User Management, Repository-Service-Router architecture.

Core Modules

Crew Management, Lobby Management, Train Management, Duty Management, Excel Import, Dashboard APIs, Reports, Audit Logs, CMS Automation.

Database Tables

Users, Crew, Lobby, Train, Duty, Import Logs, Audit Logs.

Backend Architecture

Each module follows: Model → Schema → Repository → Service → API Router.

User Roles

Administrator, Controller, Supervisor, Lobby Operator, Viewer.

Roadmap

1. Finalize Database Schema
2. Crew Module
3. Lobby Module
4. Train Module
5. Duty Module
6. Excel Import
7. Dashboard
8. Reports
9. Audit Logs
10. CMS Automation
11. Frontend
12. Testing & Deployment

Success Metrics

Secure authentication, historical data preservation, fast Excel imports, dashboard updates after import, role-based access, scalable architecture.